Lecture Notes: Introduction to Stochastic Processes (MATH 5652)

By Kevin D. Stubbs

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Chapter 0: Introduction

§0.1 Why Stochastic Processes?

The world is a complicated place and no matter how hard we try, some things are purely a matter of chance. While probability theory studies randomness in general, stochastic processes try and understand how random evolves with time. Some of the reasons people study stochastic processes include:

1. To understand the physical world,
2. To predict the future given uncertainty,
3. To design fast algorithms for extremely difficult problems.

§0.2 Examples

0.2.1 Brownian Motion

In 1827, Robert Brown, an English botanist, started looking at grains of pollen suspended in water under a microscope. What he found was somewhat surprising, the grains of pollen seemed to randomly jiggle (click here for a video of pollen grains vibrating). Brown later tried the same experiment with small particles glass and rock dust so ruled out the motion was related to life.

A mathematical understanding of Brown's observations, now called Brownian motion, was eventually provided by Albert Einstein in 1905 and was instrumental in the wide acceptance of atomic and molecular theory. Per Einstein's explanation, Brownian motion happens because the water molecules are moving randomly and hit the pollen grain at random times which causes the vibrations (click here for an animation of Brownian motion).

0.2.2 Medical Epidemiology

Since the transmission of infectious diseases is inherently uncertain, stochastic processes play an important role in modeling the incidence and spread of disease. By incorporating stochastic processes into disease models, we can account for random fluctuations arising from demographic variability, environmental differences, and discrete transmission events. This provides greater insight into how diseases spread and affect different segments of the population. More broadly, stochastic processes gives the right mathematical framework for quantifying uncertainty and characterizing the range of possible outcomes in disease dynamics.

0.2.3 Diffusion Models for Image Generation

In recent years, we've seen an explosion in the power of artificial intelligence (AI) in a wide variety of tasks. While generally people use "AI" as a catchall term, the underlying approach depends a lot on what problem you're looking at.

For image generation, one of the current best approaches are a family of methods known as "diffusion methods". These methods work by first taking a set of images and slowly by add random noise to the image. After enough steps of adding noise, you then determine then undo the noise you added at each step. Once you train the diffusion model on enough data, you learn two things: (1) a "forward map" which takes images to noise, and (2) a "backwards map" which takes noise to images. You can use the backwards map to generate new images from noise. Both the forwards map and the backwards map are modeled with stochastic processes.

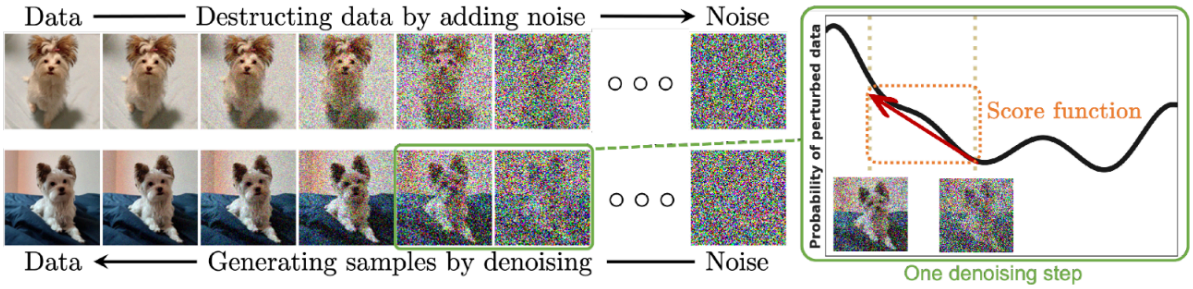


Figure 1: An illustration of how a diffusion model works for image generation. Taken from [1]. In the first row, an image of a dog has noise slowly added to it until it looks completely random. In the second row, denoising is performed to random noise to change the noise into a picture of a different dog.

§0.3 A Review of Probability Theory

0.3.1 Axioms of Probability

We start by recalling the definition of a probability space

Definition 1

A **probability space** is defined by three pieces of information:

1. *Sample space* (denoted Ω) which contains all possible outcomes,
2. *Event space* (denoted $\mathcal{F}$) a set of (reasonable) subsets the sample space, and
3. *Probability function* (denoted $\mathbb{P}$) which assigns each event a probability of occurring.

A probability function is required to satisfy three axioms:

1. If A is an event, then $0 \leq \mathbb{P}(A) \leq 1$.
2. The sum over all outcomes equals 1: $\sum_{s \in S} \mathbb{P}(s) = 1$

3. If A and B are mutually exclusive events, then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.

One of the key concepts in probability theory is *conditional probability*:

Definition 2

Given two events A, B where $\mathbb{P}(B) \neq 0$, the **conditional probability** of A given B is defined:

$$\mathbb{P}(A|B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \tag{1}$$

Theorem 1 (Bayes' Theorem)

Given two events A, B where $\mathbb{P}(B) \neq 0$ we have

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)} \tag{2}$$

Definition 3

We say two events are **independent** if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. Note that this implies $\mathbb{P}(A|B) = \mathbb{P}(A)$ so long as $\mathbb{P}(B) \neq 0$.

Some useful facts which follow from the axioms:

- $\mathbb{P}(A \cap B) \leq \mathbb{P}(A)$ (restriction makes things less likely).
- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$ (mutually exclusive means $\mathbb{P}(A \cap B) = 0$).
- $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$ (the probability A does NOT happen equals 1 minus the probability A does happen).
- If $\mathbb{P}(A) = 1$, then we say the event is certain (or occurs almost surely).
- $\mathbb{P}(\varnothing) = 0$ (the probability nothing happens is zero).

0.3.2 Random Variables

A random variable X is a function which assigns values to random events. Discrete random variables are described by their probability mass function (pmf) and continuous random variables are described by their probability density function (pdf). For example, if X is a random variable representing the roll of a 6-sided die then $X \in \{1, 2, 3, 4, 5, 6\}$ and for a fair die this random variable has pmf $p_X(x) = \mathbb{P}(X = x) = \frac{1}{6}$.

Some important definitions for discrete random variables:

$$\begin{aligned} \mathbb{E}(X) &:= \sum_x x\mathbb{P}(X = x) && \text{expected value of } X \\ \mathbb{E}(f(X)) &:= \sum_x f(x)\mathbb{P}(X = x) && \text{expected value of } f(X) \\ \mathrm{Var}(X) &:= \mathbb{E}(X^2) - \mathbb{E}(X)^2 && \text{variance of } X \end{aligned} \tag{3}$$

Chapter 1: Discrete Time Markov Chains

Markov chains are named after Andrey Markov, a Russian mathematician who did pioneering work in probability and stochastic processes. Discrete time Markov chains are the basis for Markov Chain Monte Carlo (MCMC) methods which were named one of the top 10 algorithms of the 20th century [2] and used today in a variety of applications.

§1.1 Definitions and Examples

Definition 1

A **stochastic process** is defined by a collection of random variables $\{X_t\}_{t\in T}$ all defined on the same probability space. The index set T denotes time; typical choices for T are $T := \mathbb{Z}_{(\geq 0+=\{0,1,2,...\}}$ for discrete stochastic processes and $T = [0, \infty)$ for continuous stochastic processes.

For simplicity, let's assume that each X_t takes values in a state space S which is countable. To fully specify a stochastic process, we must know it's joint distribution function for all finite sets of random variables from the set. For example, we need to know the probabilities

$$\begin{gathered}\mathbb{P}(X_0 = a) \qquad \mathbb{P}(X_{10} = b) \qquad \mathbb{P}(X_0 = c, X_{21} = d) \\ \mathbb{P}(X_0 = e, X_1 = f, X_3 = g, X_4 = h)\end{gathered} \tag{4}$$

where $a, b, ..., h \in S$. Generally speaking, we can determine these probabilities from the joint distribution function

$$\mathbb{P}(X_0 = x_0, X_1 = x_1, X_2 = x_2, ..., X_{n-1} = x_{n-1}, X_n = x_n) \quad \text{where } x_i \in S \tag{5}$$

Applying the definition of conditional probability twice, we have

$$\begin{aligned}&\mathbb{P}(X_0 = x_0, X_1 = x_1, X_2 = x_2) \\ &\quad = \mathbb{P}(X_0 = x_0)\mathbb{P}(X_1 = x_1, X_2 = x_2 | X_0 = x_0) \\ &\quad = \mathbb{P}(X_0 = x_0)\mathbb{P}(X_1 = x_1 | X_0 = x_0)\mathbb{P}(X_2 = x_2 | X_1 = x_1, X_0 = x_0)\end{aligned} \tag{6}$$

Since X_t represents the state at time t, the probability $\mathbb{P}(X_1 = x_1 | X_0 = x_0)$ has the interpretation:

If x_0 happened at $t = 0$, what is the probability that x_1 happens at $t = 1$

and similarly for $\mathbb{P}(X_2 = x_2 | X_1 = x_1, X_0 = x_0)$.

In general, we can write

$$\begin{aligned}&\mathbb{P}(X_0 = x_0, X_1 = x_1, X_2 = x_2, ..., X_{n-1} = x_{n-1}, X_n = x_n) \\ &\quad = P(X_0 = x_0)P(X_1 = x_1 | X_0 = x_0) \cdots P(X_n = x_n | X_{n-1} = x_{n-1}, \cdots, X_0 = x_0)\end{aligned} \tag{7}$$

Calculating the conditional probability $P(X_n = x_n | X_{n-1} = x_{n-1}, \cdots, X_0 = x_0)$ can be very complicated because it depends an ALL the previous times $\{X_0, \cdots, X_{n-1}\}$; this is not the case for Markov chains.

Definition 2

Let $\{X_n\}_{n\geq 0}$ be a a discrete stochastic processes taking values in a countable sample space S. We say $\{X_n\}_{n\geq 0}$ is a **discrete Markov chain** if for all $n \geq 0$ and all $x_1, \cdots, x_{n+1} \in S$:

$$\mathbb{P}(X_{n+1} = x_{n+1} | X_n = x_n, \cdots, X_0 = x_0) = \mathbb{P}(X_{n+1} = x_{n+1} | X_n = x_n). \tag{8}$$

Equation 8 is referred to as the "Markov property".

Another name for the Markov property used in engineering is "memorylessness". Given all of the previous states $X_0, \cdots, X_n$ to determine the conditional distribution of the next state, X_{n+1}, only the current state X_n matters because the system has forgotten about all of the states before X_n.

In this course, we will only consider Markov chains which are "time homogeneous", meaning that

$$\mathbb{P}(X_{n+1} = x | X_n = y) = \mathbb{P}(X_1 = x | X_0 = y) = p(y, x) \quad \text{for all } x, y \in S \tag{9}$$

The function $p(y, x)$ is a conditional probability so it satisfies

1. $p(y, x) \geq 0$ for all $x, y \in S$, and
2. $\sum_{x\in S} p(y, x) = \sum_{x\in S} \mathbb{P}(X_1 = x | X_0 = y) = 1$ for all $y \in S$

The function $p(y, x)$ defines a matrix P with positive entries and whose rows sum to 1. Importantly, $\sum_{y\in S} p(y, x) = \sum_{y\in S} \mathbb{P}(X_1 = x | X_0 = y)$ which does not necessarily equal 1.

We'll now go through a couple of examples of Markov chains:

Example 1 (i.i.d. random variables)

Suppose that $\{X_n\}_{n\geq 0}$ are independent, identically distributed (i.i.d.) random variables, then by independence

$$\begin{aligned} \mathbb{P}(X_{n+1} = x_{n+1} | X_n = x_n, ..., X_1 = x_1) &= \mathbb{P}(X_{n+1} = x_{n+1}) \\ &= \mathbb{P}(X_{n+1} = x_{n+1} | X_n = x_n) \end{aligned} \tag{10}$$

So this defines a Markov chain. The transition function is

$$p(y, x) = \mathbb{P}(X_1 = x | X_0 = y) = \mathbb{P}(X_1 = x) = p_X(x) \tag{11}$$

where $p_X(x)$ is the p.m.f. of X_1.

Example 2 (Weather Chain)

Let X_n be the weather on day n in sunny California, which we assume is either: $1 =$ *rainy*, or $2 =$ *sunny*. Even though the weather is not exactly a Markov chain, we can propose a Markov chain model for the weather by writing down a transition probability matrix

$$P = \begin{pmatrix} .9 & .1 \\ .8 & .2 \end{pmatrix} \tag{12}$$

Q: In the long run, what percentage of days are sunny?

Example 3 (Binary Erasure Channel)

Consider a communications system which attempts to transmit the bits 0 and 1. Each bit transmitted must pass through multiple stages and each stage there is a probability $\theta \in (0,1)$ that the bit becomes erased (denoted ε) Let X_n denote the digit exiting the n-th stage. Then $\{X_n\}_{n\geq 0}$ is a two-state MC with state space $S = \{0, 1, \varepsilon\}$ and a transition matrix

$$P = \begin{pmatrix} 1-\theta & 0 & \theta \\ 0 & 1-\theta & \theta \\ 0 & 0 & 1 \end{pmatrix} \tag{13}$$

Q: After n stages, what's the probability the bit hasn't been erased? What happens as $n \longrightarrow \infty$?

Example 4 (Ising Model on Two Spins)

A simple model for magnetism was studied by Ernst Ising in his PhD thesis in 1924; this model is now called the Ising model. The Ising model was completely solved in 1D by Ising in his thesis and about 20 years later, Lars Onsagar solved the Ising model in 2D. Some aspects of the Ising model in 3D remain unsolved to this day.

For two spins, Ising model can be described by a stochastic process $\{X_n\}_{n\geq 0}$ with state space $S = \{\uparrow\uparrow, \downarrow\uparrow, \uparrow\downarrow, \downarrow\downarrow\}$ and transition matrix

$$P = \begin{pmatrix} p & (1-p)/2 & (1-p)/2 & \\ p/2 & 1-p & 0 & p/2 \\ p/2 & 0 & 1-p & p/2 \\ 0 & (1-p)/2 & (1-p)/2 & p \end{pmatrix} \tag{14}$$

The parameter $p \in [0,1]$ depends on the ambient temperature. At zero temperature $p = 0$ whereas at high temperatures $p \approx \frac{1}{2}$.

Q: In the long run, does the material favor ferromagnetic states, $\{\uparrow\uparrow, \downarrow\downarrow\}$, or anti-ferromagnetic states, $\{\uparrow\downarrow, \downarrow\uparrow\}$? How does this depend on p?

§1.2 Multistep Transition Probabilities

Now that we know a bit about how Markov chains are defined, it's time to start looking into how to calculate with them. Our running example, will be the sunny/rainy chain from the previous section which is defined by the transition matrix:

$$P = \begin{pmatrix} .9 & .1 \\ .8 & .2 \end{pmatrix} \tag{15}$$

We start by recalling three equivalent ways of specifying a Markov chain

Conditional Probability	**Transition Function**	**Transition Matrix**
$\mathbb{P}(X_{n+1} = x \mid X_n = y)$	$p(y, x)$	$P = \begin{pmatrix} .9 & .1 \\ .8 & .2 \end{pmatrix}$

You should be comfortable with converting between these different ways of specifying Markov chains.

Relating Conditional Probability and the Transition Function

For time homogeneous Markov chains (see Equation 9 above), the conditional probability doesn't depend on the time n, that is:

$$\begin{aligned} \mathbb{P}(X_{n+1} = x \mid X_n = y) &= \mathbb{P}(X_n = x \mid X_{n-1} = y) \\ &= \mathbb{P}(X_{n-1} = x \mid X_{n-2} = y) \\ &= \qquad\qquad \vdots \\ &= \mathbb{P}(X_1 = x \mid X_0 = y) \end{aligned} \tag{16}$$

There are many real world systems which satisfy the Markov property but are time inhomogenous but we won't consider them in this class. Because of this homogeneity in time, the conditional probability satisfies:

$$\mathbb{P}(X_{n+1} = x \mid X_n = y) = p(y, x) \quad \text{for all } n \geq 0 \tag{17}$$

It's important to note that the order of the variables gets reversed when defining the transition function. The mnemonic to remember is that $\mathbb{P}(X_{n+1} = x \mid X_n = y)$ is the probability of starting at y and going to x so $p(y, x)$ "$= p(y \to x)$".

Relating the Transition Function and the Transition Matrix

The entries of the transition matrix correspond to the rows/columns of the transition matrix

$$p(\underbrace{y}_{\text{row}}, \overbrace{x}^{\text{column}}) = P_{y,x} \tag{18}$$

For the sunny/rainy transition matrix

$$p(1,1) = .9 \qquad p(2,2) = .2 \qquad p(1,2) = .1 \qquad p(2,1) = .8. \tag{19}$$

1.2.1 Predicting the Future

A natural question which arises in studying Markov chains is what the probability of a certain outcome

Question 1

Suppose that it's sunny (1) today, what's the probability it'll be rainy (2) tomorrow?

Answer

By definition, the probability we're looking for is

$$\mathbb{P}(X_1 = 2 | X_0 = 1) = p(1,2) = .1 \tag{20}$$

Easy!

Now let's ask a more complicated question

Question 2

Suppose that it's sunny (1) today, what's the probability it'll be rainy (2) **two** days from now?

In this case the probability we're looking for is $\mathbb{P}(X_2 = 2 | X_0 = 1)$. Since X_2 is two steps in the future, we cannot directly use the transition matrix (which only has one step). Instead, we use the law of total probability which is proven in Section 2.4.

Answer

By the law of total probability

$$\begin{aligned}
&\mathbb{P}(X_2 = 2 | X_0 = 1) \\
&= \sum_{z \in \{1,2\}} \mathbb{P}(X_2 = 2 | X_1 = z, X_0 = 1) \mathbb{P}(X_1 = z | X_0 = 1) \\
&= \sum_{z \in \{1,2\}} \mathbb{P}(X_2 = 2 | X_1 = z) \mathbb{P}(X_1 = z | X_0 = 1) \qquad \text{(Markov property)} \\
&= \mathbb{P}(X_2 = 2 | X_1 = 1) \mathbb{P}(X_1 = 1 | X_0 = 1) + \mathbb{P}(X_2 = 2 | X_1 = 2) \mathbb{P}(X_1 = 2 | X_0 = 1) \\
&= p(1,2)p(1,1) + p(2,2)p(1,2) \\
&= (.1)(.9) + (.2)(.1) = \boxed{.11}
\end{aligned}$$

Question 3

Suppose that it's sunny (1) today, what's the probability it'll be rainy (2) **three** days from now?

Answer

We're going to approach this same way as the two days, by applying the law of total probability twice.

$$\begin{aligned}
&\mathbb{P}(X_3 = 2|X_0 = 1) \\
&\quad = \sum_{z_1 \in \{1,2\}} \mathbb{P}(X_3 = 2|X_1 = z_1, X_0 = 1)\mathbb{P}(X_1 = z_1|X_0 = 1) \\
&\quad = \sum_{z_1 \in \{1,2\}} \mathbb{P}(X_3 = 2|X_1 = z_1, X_0 = 1)p(1, z_1)
\end{aligned}$$

where in the second line, we've used that since $\{X_n\}_{n\geq 0}$ is a time homogeneous Markov chain $\mathbb{P}(X_1 = z_1|X_0 = 1) = p(1, z_1)$.

To simplify the remaining $P(X_3 = 2| \cdots)$ we can apply the law of total probability again:

$$\begin{aligned}
&\mathbb{P}(X_3 = 2|X_1 = z_1, X_0 = 1) \\
&\quad = \sum_{z_2 \in \{1,2\}} \mathbb{P}(X_3 = 2|X_2 = z_2, X_1 = z_1, X_0 = 1)\mathbb{P}(X_2 = z_2|X_1 = z_1, X_0 = 1) \\
&\quad = \sum_{z_2 \in \{1,2\}} \mathbb{P}(X_3 = 2|X_2 = z_2)\mathbb{P}(X_2 = z_2|X_1 = z_1) \\
&\quad = \sum_{z_2 \in \{1,2\}} p(z_2, 2)p(z_1, z_2)
\end{aligned}$$

It's important to note that *first line* is true for any stochastic process, but the *second line* (which removed the $X_1 = z_1, X_0 = 1$ from the first $\mathbb{P}$ and $X_0 = 1$ from the second $\mathbb{P}$), is only because $\{X_n\}_{n\geq 0}$ forms a Markov chain (and the third line is because of time homogeneity). Putting it all together, we have

$$\begin{aligned}
\mathbb{P}(X_3 = 2|X_1 = z_1, X_0 = 1) &= \sum_{z_1 \in \{1,2\}} \left(\sum_{z_2 \in \{1,2\}} p(z_2, 2)p(z_1, z_2) \right) p(1, z_1) \\
&= \sum_{z_1, z_2 \in \{1,2\}} p(z_2, 2)p(z_1, z_2)p(1, z_1) \\
&= \sum_{z_1, z_2 \in \{1,2\}} p(1, z_1)p(z_1, z_2)p(z_2, 2).
\end{aligned}$$

For those who know linear algebra, this last line is actually equal to the $(1, 2)$ entry of the matrix $P^3 = P \cdot P \cdot P$; proving this fact will be our next step.

1.2.2 General k-step transition formula

Let A be a $n \times p$ and B be a $p \times m$ matrix, the product of AB is defined to be

$$[AB]_{ij} = \sum_{k=1}^{p} [A]_{ik}[B]_{kj}. \tag{21}$$

Theorem 1

Let $\{X_n\}_{n\geq 0}$ be a Markov chain with transition matrix P.

$$\mathbb{P}(X_{n+k} = x | X_n = y) = \left[P^k\right]_{y,x} \tag{22}$$

Proof: We'll prove this by mathematical induction. Let's start by thinking about $k = 1$

$$\mathbb{P}(X_{n+1} = x | X_n = y) = p(y, x) = [P]_{y,x} \qquad \text{(definition)} \tag{23}$$

As for $k = 2$, by law of total probability

$$\begin{aligned}
&\mathbb{P}(X_{n+2} = x | X_n = y) \\
&\quad = \sum_{z\in S} \mathbb{P}(X_{n+2} = x | X_{n+1} = z, X_n = y)\mathbb{P}(X_{n+1} = z | X_n = y) \\
&\quad = \sum_{z\in S} \mathbb{P}(X_{n+2} = x | X_{n+1} = z)\mathbb{P}(X_{n+1} = z | X_n = y) \quad \text{(Markov property)} \\
&\quad = \sum_{z\in S} [P]_{z,x}[P]_{y,z} \qquad \text{(definition of the transition matrix)} \\
&\quad = \sum_{z\in S} [P]_{y,z}[P]_{z,x} \\
&\quad = \left[P^2\right]_{y,x} \qquad \text{(definition of matrix multiplication)}
\end{aligned} \tag{24}$$

For $k = 3$, the steps are almost the same:

$$\begin{aligned}
&\mathbb{P}(X_{n+3} = x | X_n = y) \\
&\quad = \sum_{z\in S} \mathbb{P}(X_{n+3} = x | X_{n+2} = z, X_n = y)\mathbb{P}(X_{n+2} = z | X_n = y) \\
&\quad = \sum_{z\in S} \mathbb{P}(X_{n+3} = x | X_{n+1} = z)\mathbb{P}(X_{n+2} = z | X_n = y) \\
&\quad = \sum_{z\in S} \mathbb{P}(X_{n+2} = z | X_n = y)[P]_{z,x}
\end{aligned} \tag{25}$$

Now notice that we already know that

$$\mathbb{P}(X_{n+2} = z | X_n = y) = \left[P^2\right]_{y,z} \tag{26}$$

from the previous step. Therefore,

$$\mathbb{P}(X_{n+3} = x | X_n = y) = \sum_{z \in S} \left[P^2\right]_{y,z} [P]_{z,x} = \left[P^3\right]_{y,x}. \tag{27}$$

The key trick here is that we used $k = 2$ to prove the formula for $k = 3$. Therefore, for general k

$$\begin{aligned}
&\mathbb{P}(X_{n+k} = x | X_n = y) \\
&\quad = \sum_{z \in S} \mathbb{P}(X_{n+k} = x | X_{n+k-1} = z, X_n = y) \mathbb{P}(X_{n+k-1} = z | X_n = y) \\
&\quad = \sum_{z \in S} \mathbb{P}(X_{n+k-1} = z | X_n = y) [P]_{z,x} \\
&\quad = \sum_{z \in S} \left[P^{k-1}\right]_{y,z} [P]_{z,x} \quad \text{(since we already proved for the previous step)} \\
&\quad = \left[P^k\right]_{y,x}.
\end{aligned} \tag{28}$$

This completes the proof.

■

1.2.3 Chapman–Kolmogorov Equation and Probabilistic Interpretation

There are a lot of different approaches to proving Equation 22 and above we used the one which ties most directly to linear algebra. Another way of interpreting this result is through the following equation which is know as the Chapman–Kolmogorov equation:

$$\mathbb{P}(X_{m+n} = x | X_0 = y) = \sum_{z \in S} \left[P^m\right]_{y,z} \left[P^n\right]_{z,x} \tag{29}$$

The way you can see this from Equation 22 is by noting that

$$\mathbb{P}(X_{m+n} = x | X_0 = y) = \left[P^{m+n}\right]_{y,x} = \sum_{z \in S} \left[P^m\right]_{y,z} \left[P^n\right]_{z,x} \tag{30}$$

where the last equality is because of the definition of matrix multiplication. The Chapman–Kolmogorov equation has the following interpretation

Chapman–Kolmogorov Interpretation For a Markov chain, the following are equivalent:

1. We start the chain at y, let it run for $m + n$ steps, and see what the probability we end up at x is.
2. We start the chain at y, let it run m steps and suppose we end up at z. Then we start the chain at z, let it run for n steps, and see what the probability we end up at x is.

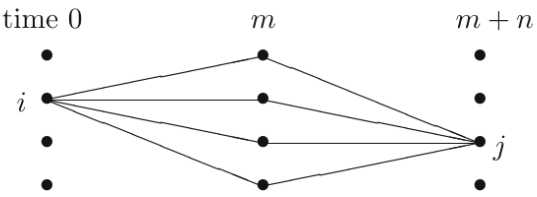


Figure 2: Illustration from the book [3], of the Chapman–Kolmogorov Equation. Here, we start at i and end at j (instead of y and x).

§1.3 Classification of States

Definition 3

From now, given a Markov chain $\{X_n\}_{n\geq 0}$ and for any event A and any $x \in S$, we define

$$\mathbb{P}_x(A) := \mathbb{P}(A|X_0 = x) \quad \text{and} \quad \mathbb{E}_x(A) := \mathbb{E}(A|X_0 = x) \tag{31}$$

So $\mathbb{P}_x(A)$ and $\mathbb{E}_x(A)$ are the probability and expectation of A happening given the chain starts at x.

In Markov chains, there are two classes of states: **recurrent** and **transient**.

Definition 4

Given a Markov chain $\{X_n\}_{n\geq 0}$ on a state space S, we say that:

$$\begin{aligned} x \text{ is } \textbf{recurrent} &\iff \mathbb{P}_x(\text{we eventually return to } x) = 1, \\ x \text{ is } \textbf{transient} &\iff \mathbb{P}_x(\text{we eventually return to } x) < 1. \end{aligned} \tag{32}$$

To make sense of recurrence/transience mathematically, we need to first talk about *stopping times* and the *strong Markov property*.

Definition 5

Given a stochastic process $\{X_n\}_{n\geq 0}$ a **stopping time** is any time T so that we can determine whether or not the event $\{T = n\}$ happened by looking at the values of $\{X_0, \cdots, X_n\}$.

One important stopping time is the *time of first return*:

Example 5

Given a stochastic process $\{X_n\}_{n\geq 0}$ we define the **time of first return** (to x) as

$$T_x := \min\{n \geq 1 : X_n = x\}. \tag{33}$$

Why is this a stopping time? Consider the following

- $\{T_x = 1\} = \{X_0 = x, X_1 = x\}$
- $\{T_x = 2\} = \{X_0 = x, X_1 \neq x, X_2 = x\}$ because T_x is the FIRST return to x,
- $\{T_x = 3\} = \{X_0 = x, X_1 \neq x, X_2 \neq x, X_3 = x\}$,
- ...
- $\{T_x = n\} = \{X_0 = x, X_1 \neq x, \cdots, X_{n-1} \neq x, X_n = x\}$.

For each n, we can determine whether $\{T_x = n\}$ happened or not by looking at the values of $\{X_0, \cdots, X_n\}$.

The reason stopping times are so important is because of the *strong Markov property*:

Theorem 2 (Strong Markov Property, informal)

Let T be a stopping time for a Markov chain $\{X_n\}_{n\geq 0}$. Given $\{T = n\}$ and $\{X_T = x\}$ any other information about the previous times $\{X_0, \cdots, X_{T-1}\}$ is irrelevant for predicting the future states X_{T+k} for any $k \geq 0$. In particular, X_{T+k} behaves like a Markov chain with initial state x.

All Markov chains satisfy the strong Markov property so in particular

$$\begin{aligned} &\mathbb{P}(X_{T+1} = x | X_T = y, T = n) \\ &\quad = \mathbb{P}(X_{n+1} = x | X_n = y) \quad \text{when } \{X_n\}_{n\geq 0} \text{ is a Markov chain} \\ &\quad = p(y, x) \end{aligned} \tag{34}$$

Caution For general stochastic processes you *cannot* drop the $T = n$. For any stochastic process $\{Y_n\}_{n\geq 0}$ you have

$$\mathbb{P}(Y_{T+1} = y | Y_T = x, T = n) = \mathbb{P}(Y_{n+1} = y | Y_n = x, T = n) \tag{35}$$

however in general $\mathbb{P}(Y_{n+1} = y | Y_n = x, T = n) \neq \mathbb{P}(Y_{n+1} = y | Y_n = x)$.

Proving that Markov chains satisfy the strong Markov property in full generality is beyond the scope of this course, but we'll prove it in a simple case which gives the flavor of the argument. Suppose that T is a stopping time and consider the probability

$$\mathbb{P}(X_{T+1} = x | X_T = y, T = n) \tag{36}$$

Since T is a stopping time, we can write

$$\{T = n\} = \bigcup_{(z_0, \cdots, z_n) \in V_n} \{X_0 = z_0, X_1 = z_1, \cdots, X_n = z_n\} \tag{37}$$

where V_n is the set of sequences of steps where $\{T = n\}$ is happened. By the law of total probability:

$$\begin{aligned}
&\mathbb{P}(X_{T+1} = x, X_T = y, T = n) \\
&= \sum_{(z_0,\cdots,z_n)\in V_n} \mathbb{P}(X_{n+1} = x, X_n = y, X_0 = z_0, \cdots, X_n = z_n) \\
&= \sum_{(z_0,\cdots,z_n)\in V_n} \mathbb{P}(X_{n+1} = x | X_n = y, X_0 = z_0, \cdots, X_n = z_n)\mathbb{P}(X_0 = z_0, \cdots, X_n = z_n, X_n = y) \\
&= \sum_{(z_0,\cdots,z_n)\in V_n} \mathbb{P}(X_{n+1} = x | X_n = y)\mathbb{P}(X_0 = z_0, \cdots, X_n = z_n, X_n = y) \\
&= \mathbb{P}(X_{n+1} = x | X_n = y)\left(\sum_{(z_0,\cdots,z_n)\in V_n} \mathbb{P}(X_0 = z_0, \cdots, X_n = z_n, X_n = y)\right) \\
&= \mathbb{P}(X_{n+1} = x | X_n = y)\mathbb{P}(X_n = y, T = n)
\end{aligned} \tag{38}$$

Therefore,

$$\begin{aligned}
\mathbb{P}(X_{n+1} = x | X_n = y) &= \frac{\mathbb{P}(X_{T+1} = x, X_T = y, T = n)}{\mathbb{P}(X_n = y, T = n)} \\
&= \mathbb{P}(X_{T+1} = x | X_T = y, T = n).
\end{aligned} \tag{39}$$

1.3.1 The Time of First Return, Recurrence, and Transience

Definition 6

For all $x, y \in S$, the probability we eventually go to y given we started at x is given by:

$$\rho_{xy} := \mathbb{P}_x\big(T_y < \infty\big) \tag{40}$$

Note that

$$\mathbb{P}_x(\text{we eventually return to } x) = \mathbb{P}(T_x < \infty) = \rho_{xx}. \tag{41}$$

We can also consider the time of kth return:

$$T_x^{(k)} := \min\Big\{n \geq T_x^{(k-1)} : X_n = x\Big\}. \tag{42}$$

By the strong Markov property

$$\mathbb{P}\Big(T_x^{(k)} < \infty\Big) = \big(\mathbb{P}(T_x < \infty)\big)^k = \rho_{xx}^k \tag{43}$$

Before proving this equality, we'll note that fact leads to two cases:

1. If $\rho_{xx} < 1$ then

$$\mathbb{P}\Big(T_x^{(k)} < \infty\Big) = \rho_{xx}^k \to 0 \text{ as } k \to \infty \quad (\textbf{transience}) \tag{44}$$

2. If $\rho_{xx} = 1$ then

$$\mathbb{P}\left(T_x^{(k)} < \infty\right) = 1 \text{ for all } k \quad \textbf{(recurrence)}. \tag{45}$$

Proof: We'll prove this for $k = 2$, the general result is by mathematical induction. The two events

$$\left\{T_x^{(1)} < \infty\right\} \cup \left\{T_x^{(1)} = \infty\right\} \tag{46}$$

form a partition so by law of total probability

$$\mathbb{P}_x\left(T_x^{(2)} < \infty\right) = \mathbb{P}_x\left(T_x^{(2)} < \infty, T_x < \infty\right) + \mathbb{P}_x\left(T_x^{(2)} < \infty, T_x = \infty\right) \tag{47}$$

But $\mathbb{P}_x\left(T_x^{(2)} < \infty, T_x = \infty\right) = 0$ because if the event $\{T_x = \infty\}$ happens, there's no way $\left\{T_x^{(2)} < \infty\right\}$ can happen. Therefore,

$$\begin{aligned} \mathbb{P}_x\left(T_x^{(2)} < \infty\right) &= \mathbb{P}_x\left(T_x^{(2)} < \infty, T_x < \infty\right) \\ &= \sum_{n=1}^{\infty} \mathbb{P}_x\left(T_x^{(2)} < \infty | T_x = n\right) \mathbb{P}_x(T_x = n) \end{aligned} \tag{48}$$

But $\mathbb{P}_x\left(T_x^{(2)} < \infty | T_x = n\right) = \mathbb{P}_x(T_x < \infty) = \rho_{xx}$ by the strong Markov property. Substituting this back into the previous expression

$$\mathbb{P}_x\left(T_x^{(2)} < \infty\right) = \rho_{xx} \sum_{n=1}^{\infty} \mathbb{P}_x(T_x = n) = (\rho_{xx})^2 \tag{49}$$

■

Example 6 (Gambler's Ruin)

KDS: Insert example for worksheet 2

Definition 7

We say that x **communicates with** y (denoted $x \to y$) if

$$\rho_{xy} = \mathbb{P}_x\left(T_y < \infty\right) > 0 \tag{50}$$

That is, the probability that we eventually reach y starting at x is non-zero.

Example 7

KDS: Which states communicate in gambler's example?

Example 8

KDS: New example with transition matrix

Lemma 1

If $x \to y$ and $y \to z$, then $x \to z$.

Proof: By definition

$$\mathbb{P}_x\big(T_y < \infty\big) = \sum_{m=0}^{\infty} \mathbb{P}_x\big(T_y = m\big) = \tag{51}$$

■

Theorem 3

If x is recurrent and $\rho_{yx} > 0$, then $\rho_{yx} = 1$.

§1.4 Proof of the Law of Total Probability

We'll start with the abstract statement of the law of total probability

Theorem 4 (Law of Total Probability)

Let A be any event and let $\{B_i\}_i$ be a (countable) partition of probability space Ω, meaning it satisfies

$$\bigcup_i B_i = \Omega \qquad \text{and} \qquad \mathbb{P}\big(B_i \cap B_j\big) = 0 \text{ for all } i, j \tag{52}$$

then

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A|B_i)\mathbb{P}(B_i) = \sum_i \mathbb{P}(A \cap B_i) \tag{53}$$

Additionally, if C is any event with $\mathbb{P}(C \neq 0)$ then

$$\mathbb{P}(A|C) = \sum_i \mathbb{P}(A|C \cap B_i)\mathbb{P}(B_i|C) \tag{54}$$

Often times in this class, we'll have a random variable X which has possible outcomes, S. We'll apply the law of total probability with the partition $\{\{X = s\} : s \in S\}$ (why is this a partition?). In practice, applying the law of total probability gives you equalities like

$$\begin{aligned} \mathbb{P}(Y = y) &= \sum_{s \in S} \mathbb{P}(Y = y|X = s)\mathbb{P}(X = s), \quad \text{or} \\ \mathbb{P}(Y = y|Z = z) &= \sum_{s \in S} \mathbb{P}(Y = y|Z = z, X = s)\mathbb{P}(X = s|Z = z) \end{aligned} \tag{55}$$

Proof: The key fact underlying this proof is that if $\{E_i\}$ is a countable family of events then

$$\mathbb{P}\big(E_i \cap E_j\big) = 0 \text{ for all } i, j \implies \mathbb{P}\left(\bigcup_i E_i\right) = \sum_i \mathbb{P}(E_i) \tag{56}$$

This fact is one of the axioms of probability.

Remark 1

If E_1, E_2 are two events then this statement simplifies to

$$\mathbb{P}(E_1 \cap E_2) = 0 \implies \mathbb{P}(E_1 \cup E_2) = \mathbb{P}(E_1) + \mathbb{P}(E_2). \tag{57}$$

which can be see as a consequence of the inclusion–exclusion principle.

Since $\bigcup_i B_i = \Omega$, we have that

$$\mathbb{P}(A) = \mathbb{P}\left(A \cap \bigcup_i B_i\right) = \mathbb{P}\left(\bigcup_i (A \cap B_i)\right). \tag{58}$$

But we can define $E_i \coloneqq A \cap B_i$ and we have that

$$\begin{aligned} \mathbb{P}\big(E_i \cap E_j\big) &= \mathbb{P}\big((A \cap B_i) \cap \big(A \cap B_j\big)\big) \\ &= \mathbb{P}\big(A \cap \big(B_i \cap B_j\big)\big) \\ &\leq \mathbb{P}\big(B_i \cap B_j\big) = 0. \end{aligned} \tag{59}$$

Therefore,

$$\mathbb{P}(A) = \mathbb{P}\left(\bigcup_i (A \cap B_i)\right) = \sum_i \mathbb{P}(A \cap B_i) = \sum_i \mathbb{P}(A|B_i)\mathbb{P}(B_i). \tag{60}$$

For the second statement, we can use roughly the same steps:

$$\begin{aligned} \mathbb{P}(A|C) &= \mathbb{P}\left(\bigcup_i (A \cap B_i) \mid C\right) \\ &= \sum_i \mathbb{P}(A \cap B_i|C) \\ &= \sum_i \mathbb{P}(A|B_i \cap C)\mathbb{P}(B_i|C). \end{aligned} \tag{61}$$

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